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January 12, 1966
SUPERSEDING
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February 17, 1953

FEDERAL SPECIFICATION

GLASS (LABORATORY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the glass used in the fabrication of laboratory glassware.

1.2 Classification.

1.2.1 ***Types and classes.*** The glass shall be of the following types and classes, as specified (see 6.2):

Type I. Borosilicate glass.
Class A.
Class B.

Type II. Soda-lime glass.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. There are no specifications and standards applicable to this specification.

2.2 Other publications. The following documents for a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publications:

ASTM C 225—Methods of Test for Resistance of Glass Containers to Chemical Attack.

ASTM C 336—Method of Test for Annealing Point and Strain Point of Glass.

ASTM C 337—Method of Test for Average Linear Expansion of Glass.

ASTM C 338—Method of Test for Softening Point of Glass.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

3. REQUIREMENTS

3.1 Material. Glass shall be produced from inorganic oxides.

3.2 Physical requirements.

3.2.1 ***Softening point.*** The softening point of the glass, when tested as specified in 4.4.1, shall be within the range specified in table I for the particular type and class.

3.2.2 ***Annealing point.*** The annealing point of the glass, when tested as specified in 4.4.2, shall be within the range specified in table I for the particular type and class.

3.2.3 ***Coefficient of expansion.*** The coefficient of expansion of the glass, when tested as specified in 4.4.3, shall be within the range specified in table I for the particular type and class.

3.3 Chemical requirements.

3.3.1 ***Solubility.*** The solubility, when tested as specified in 4.4.4, shall be not greater than that specified in table I for the particular type and class.

3.3.2 ***Lead oxides.*** Lead oxides shall not be present in excess of the percentage specified in table I for the particular type and class, when tested as specified in 4.4.5.

3.3.3 ***Arsenic oxide and antimony oxide.*** Arsenic oxide and antimony oxide shall be

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not present in excess of the percentage specified in table I for the particular type and class, when tested as specified in 4.4.6.

3.3.4 *Calcium oxide.* Calcium oxide shall not be present in excess of the percentage specified in table I for the particular type and class, when tested as specified in 4.4.7.

3.3.5 *Magnesium oxide.* Magnesium oxide shall not be present in excess of the percentage specified in table I for the particular type and class, when tested as specified in 4.4.8.

3.3.6 *Zinc oxide.* Zinc oxide shall not be present in excess of the percentage specified in table I for the particular type and class, when tested as specified in 4.4.9.

TABLE I.

	Type I		Type II
	Class A	Class B	
Softening point °C.	820±10	799±10	700±10
Annealing point °C.	560±10	574±10	520±10
*Coefficient of expansion cm./cm. °C.	32.0± 1.5×10⁻⁷	50.0± 2/4×10⁻⁷	93.0± 2×10⁻⁷
Solubility (ml. N/50 H_2SO_4 per 10 grams (maximum)	1.0	1.0	9.5
Percent lead oxide (PbO) (maximum)	0.1	0.1	0.1
Percent arsenic oxide (A_2O_3) and antimony oxide (Sb_2O_3) (combined) (maximum)	0.005	0.1	0.1
Percent calcium oxide (Ca O) (maximum)	0.1		
Percent magnesium oxide (Mg O) (maximum)	0.1		
Percent zinc oxide (Zn O) (maximum)	0.1		

*Test shall be performed from 0° to 300° C.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Supplier's standard sampling plan may be used provided it is approved by the Government.

4.3 Examination. The glass shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Test methods specified in 4.4.1 through 4.4.4 shall be as indicated in ASTM Standards.

4.4.1 *Softening point.* ASTM C 338.

4.4.2 *Annealing point.* ASTM C 336.

4.4.3 *Coefficient of expansion.* ASTM C 337.

4.4.4 *Solubility.* ASTM C 225.

4.4.5 *Lead oxide.**

4.4.6 *Arsenic oxide plus antimony oxide.**

4.4.7 *Calcium oxide.**

4.4.8 *Magnesium oxide.**

4.4.9 *Zinc oxide.**

*Supplier's standard test method shall be used provided it is approved by the Government (see 6.3).

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5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.** The commodity shall be packaged and packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 **Marking.**

5.2.1 *Interior packages.* Each package, when finished, shall be marked to identify contents.

5.2.2 *Exterior containers.* Exterior containers shall be marked with name of material, quantity, name of consignee, name of consignor, and contract or purchase order number.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Number and date of this specification.

(*b*) Type and class required (see 1.2.1).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Previous approval of supplier's test method may be used as approval by the procuring activity, provided the supplier has not revised his method. The supplier may submit test method and revisions for review and approval by the procuring activity at any time.

MILITARY CUSTODIANS:

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Preparing activity:

Defense Supply Agency—DM

Review and User information is current as of the date of this document. For future coordination of changes to this document draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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